Amharic Legal Chatbot

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Overview

- Introduction
- Motivation & Problem Statement
- Data Acquisition
- Technical Architecture
- Implementation Details
- Evaluation & Performance Results
- System Demo & UI
- Challenges Overcome
- Future Roadmap

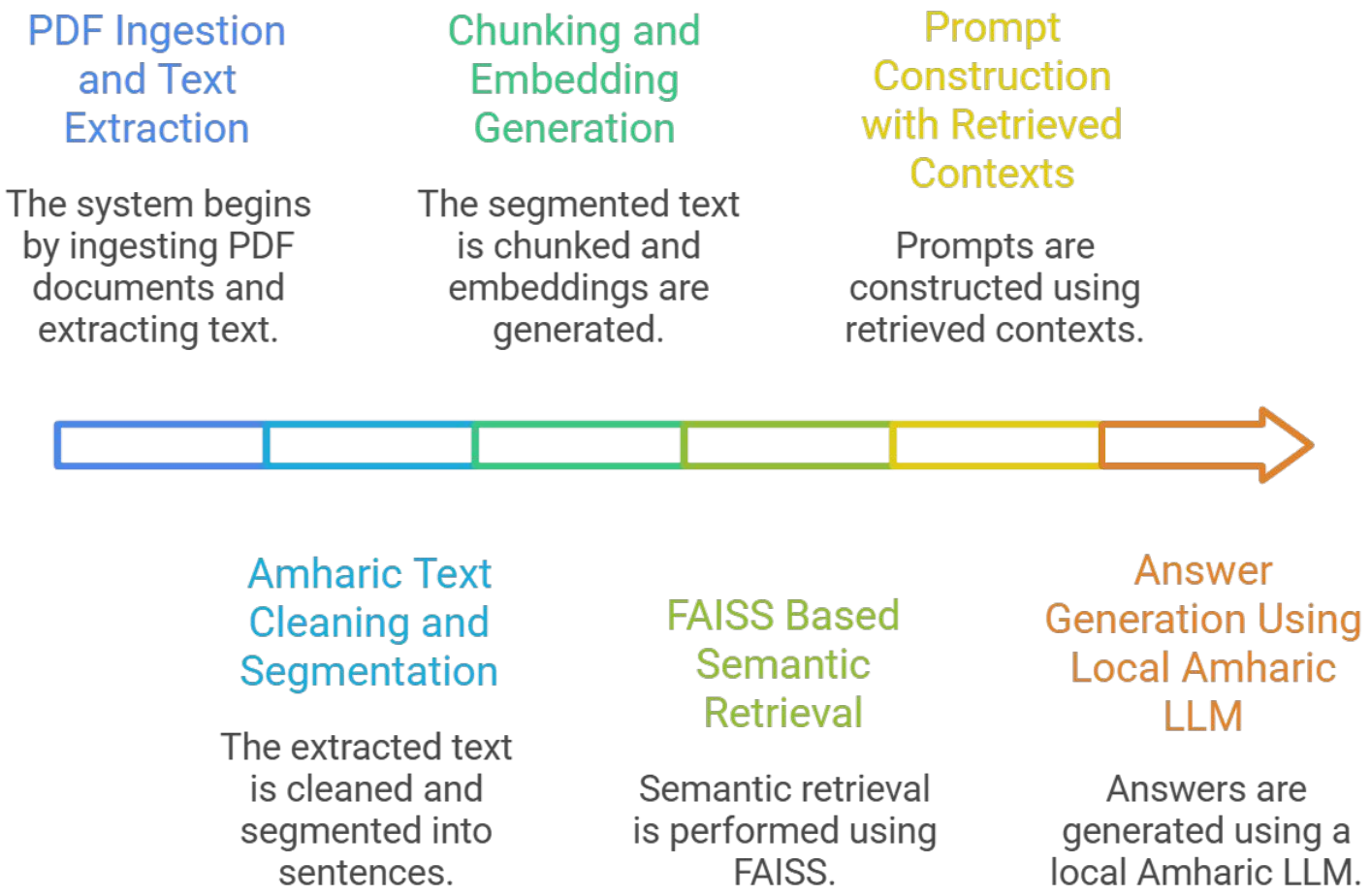
Introduction

- **Goal**: To build an AI-powered legal assistant capable of processing Amharic queries and providing grounded, **legally accurate** responses based on Ethiopian law.
- **Solution**: Developed a Retrieval-Augmented Generation (RAG) pipeline utilizing the Llama 3.2 1B Amharic and a specialized vector database of Ethiopian legal texts.
- **Impact**: Bridges the gap between complex legal texts and the public, reducing research time for professionals and increasing access to legal information for Amharic speakers.

Motivation & The Need: Why this project matters

- Domain & Language Mismatch: General purpose language models hallucinate or give unreliable legal answers.
- Access to Legal Information: underserved regions lack access to timely legal counsel. This tool provides a preliminary, low-cost legal grounded analysis.
- Efficiency for Professionals: Reduces hours of manual research through physical archives or non-searchable PDFs.
- Legal text requires grounded answers based on official documents. RAG enables answers to be generated strictly from retrieved legal sources

Overall System Architecture



Data Acquisition

Primary Sources (Official):

- Federal Negarit Gazette (Proclamations & Regulations)
- Federal Supreme Court Cassation Decisions
- Sources: justice.gov.et, fsc.gov.et, nbe.gov.et.

Secondary Sources (Analysis)

- Legal commentaries and case summaries from
  - lawethiopia.com
  - chilot.wordpress.com.

Data Scraping

For each website

- Visits the homepage
- Follows internal links only (same domain)
- Stops after 10 pages (max_pages=10)
- Collects all links that end with .pdf

For each found PDF link

- Downloads the file in chunks

Amharic detection

- Reads text from the first 3 pages of the PDF
- If not text is found, it assumes that the document is OCR
- If text exists, detects and keeps documents written in Amharic

Data Preprocessing

Data Extraction Process

- Legal documents are stored as PDF files in a predefined folder
- The system automatically locates the predefined PDF folder, scans all available PDFs
- Each PDF is opened using the PyMuPDF library
- Text is extracted page by page instead of per document
- Page level extraction enables accurate source and page referencing
- Extracted text is cleaned using Amharic specific character filtering
- Non Amharic symbols and unnecessary characters are removed
- Extra white spaces are normalized to produce clean text
- The final output is structured as page wise text with source file and page metadata

Segmentation

- Cleaned text is split into sentences using Amharic sentence boundaries
- Sentence splitting is based on Amharic punctuation such as ‘::’ and newlines
- Sentence level segmentation helps maintain semantic coherence
- Segmented sentences are later grouped into overlapping chunks for retrieval efficiency

Chunking Strategy

- Chunking groups multiple segmented sentences together
- Chunking controls the size of text passed to the embedding model and LLM
- **Chunk Size**: 200 tokens with an 80-token overlap to maintain context for small LLMs (total of around **820,000 tokens**)

The Embedding Engine, Vector Store and Indexing

Embedding Model: ***paraphrase-multilingual-MiniLM-L12-v2*** (we choose this due to its cross lingual capabilities and efficiency on CPU/GPU.

FAISS (Facebook AI Similarity Search):

- Index Type: IndexIVFFlat (Inverted File Index). Index trained dynamically based on dataset size
- Metric: Inner Product (Cosine Similarity via L2 Normalization).
- Feature: Persistent storage of metadata (Source file & Page number) for traceability.

Metadata stored separately in JSON format.

Retrieval and Prompt Engineering

Retrieval

- User query embedded using the same embedding model
- Top K most similar chunks retrieved from FAISS (Cosine similarity ranks chunks for relevance)
- **Deduplication**: Ensures unique sources are provided, keeping the highest similarity scores.

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Prompt Engineering

- Retrieved chunks are combined into a single context prompt
- Context truncated to avoid overwhelming the LLM
- Instructions guide the model to answer clearly in Amharic
- Emphasis on:
  a. Using retrieved context only
  b. Avoiding inventing extra information
  c. Producing a concise, clear, human-readable answer
- Prompt includes metadata: source file name and page number for traceability

LLM Generation Model

- The LLM model: ***rasyosef/Llama-3.2-1B-Amharic***
- Receives the prompt constructed from retrieved chunks and instructions
- Generates answers in Amharic, tailored to the legal context
- Generation parameters tuned to improve quality:
  - max_new_tokens = 256
  - temperature = 0.2 for more focused responses
  - repetition_penalty = 1.8
- Prompt echo is removed to keep output clean
- Final output is concise, contextually relevant, and traceable to source PDFs

Overall Implementation Tech Stack

- Programming Language: **Python 3**
- Environment: Google Collab and Vscode
- PDF Processing: **PyMuPDF** (fitz) for text extraction
- Text Preprocessing:
  - Regular expressions for Amharic text cleaning
  - Sentence splitting and chunking
- Embeddings: sentence-transformers/**paraphrase-multilingual-MiniLM-L12-v2**
- Vector Store / Retrieval: **FAISS** for similarity search
- LLM Generation: Hugging Face transformers pipeline with **rasyosef/Llama-3.2-1B-Amharic**

Results and Demonstration

Evaluation & Results

Objective: Measure accuracy and faithfulness of generated answers against retrieved legal text.

Metrics:

1. Faithfulness: How much the answer is supported by retrieved legal text
2. BERT Score: Semantic similarity to reference answers

Dataset Evaluation:

- Total questions: 123
- **Average Faithfulness:** 0.594
- **Average BERT Score:** 0.753
- Faithfulness range: 0.306 – 0.838
- BERT Score range: 0.631 – 0.849

Observations and Insights from the Evaluation

Observations

- Some questions show high alignment with reference text (faithfulness >0.8)
- BERTScore often slightly higher than faithfulness, indicating semantic similarity even when exact content differs
- Lower scores often occur for abstract or multi-part legal questions

Insights

- System performs well for fact based legal questions
- Challenges remain for nuanced interpretation or multi-step reasoning
- Provides actionable guidance for further fine-tuning or context expansion

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Generation Quality (Human Review):

- Automatic evaluation metrics such as like semantic similarity scores, and similar overlap based measures are commonly used to assess the quality of generated text.
- These metrics are useful because they are objective, reproducible, and scalable, allowing rapid comparison between models and configurations.
- However, such metrics have limitations, particularly in open ended and knowledge intensive tasks like legal question answering. Lexical overlap metrics often fail to capture semantic correctness, legal soundness, and contextual appropriateness

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Human Evaluation:

- Legal experts can judge whether a response is legally accurate, contextually relevant, and practically useful, aspects that cannot be reliably measured by automatic metrics alone.
- Also capture clarity of explanation, completeness of legal reasoning, and potential risks such as misleading or oversimplified legal interpretations.
- For these reasons, this project adopts a human evaluation

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- 3 Legal experts
- 20 question answer pairs given to them

Evaluation Criteria

1. Relevance: How well the answer addresses the question asked.
2. Informativeness: The amount of useful information provided in the answer.
3. Clarity: How clear and understandable the explanation is.
4. Legal Accuracy: Correctness of the answer according to legal rules and sources.
5. Completeness: Whether the answer covers all important aspects of the question.

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Likert Scale Definition

1 = Very poor

2 = Poor

3 = Fair

4 = Good

5 = Excellent

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Individual Evaluator Scores

Criteria	Legal Expert A	Legal Expert B	Legal Expert C
Relevance	4	3	4
Informativeness	3	3	4
Clarity	2	4	4
Legal Accuracy	4	4	4
Completeness	2	3	3

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Average Scores Across Evaluators

- Relevance: (4 + 3 + 4) / 3 = 3.67
- Informativeness: (3 + 3 + 4) / 3 = 3.33
- Clarity: (3 + 4 + 4) / 3 = 3.67
- Legal Accuracy: (4 + 4 + 4) / 3 = 4.00
- Completeness: (2 + 3 + 3) / 3 = 2.67

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Overall Interpretation

- The system demonstrates good relevance, indicating that retrieved and generated responses are generally aligned with the legal questions posed.
- Legal accuracy scores highest, suggesting that the system produces legally sound responses with minimal factual or doctrinal errors.
- Clarity is moderately strong, showing that answers are mostly understandable, though some responses could benefit from improved structure and phrasing in Amharic.

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Overall Interpretation Cont…

- Informativeness is fair, implying that responses provide useful information but may lack sufficient detail or supporting legal references.
- Completeness is the weakest area, indicating that answers often do not fully cover all aspects of the legal issues, such as missing conditions, exceptions, or procedural details.
- Overall, the results reflect a promising but still developing Amharic legal RAG system, with particular need for improvement in response depth and coverage.

Challenges

- Model Access Limitations: Initially attempted to use Gemini via OpenRouter, but exceeded usage limits and faced subscription restrictions
- Data Quality: older scanned government documents were difficult; required significant manual cleaning.
- Morphological Complexity: Amharic's rich morphology meant standard "exact match" searches failed. We solved this using dense vector embeddings rather than keyword search.

Conclusion

- The Amharic Legal Chatbot has proven that modern AI can be effectively adapted to **low-resource languages** and **specific local legal contexts**.
- By combining the reasoning power of Llama with a strict RAG architecture, we have created a tool that is both powerful and trustworthy.
- Faithfulness and semantic similarity indicate reasonable alignment with source documents.
- Human evaluation shows strong legal accuracy, moderate clarity and informativeness, but limited completeness.

Thank You!